

Name KEY
(print)

Total = 100 points

Please show all your work.

1. (20 pts) Find the approximate mean and standard deviation of the data set.

Sample annual salaries (in thousand of dollars) for office managers:

Salary	Frequency	MIDPOINT	CUM. FREQ	$X-M$	$(X-M)^2$
30-34	7	32	7	-10.2	104.04
35-39	10	37	17	-5.2	27.04
40-44	14	42	31	-0.2	.04
45-49	12	47	43	4.8	23.04
50-54	7	52	50	9.8	96.04

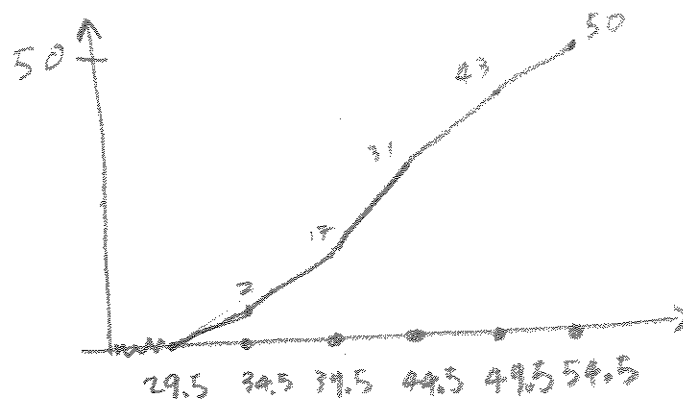
$$M = \frac{7 \times 32 + 10 \times 37 + 14 \times 42 + 12 \times 47 + 7 \times 52}{50} = \frac{2110}{50} = 42.2$$

$$SS_x = 7 \times 104.04 + 10 \times 27.04 + 14 \times .04 + 12 \times 23.04 + 7 \times 96.04 = 1948$$

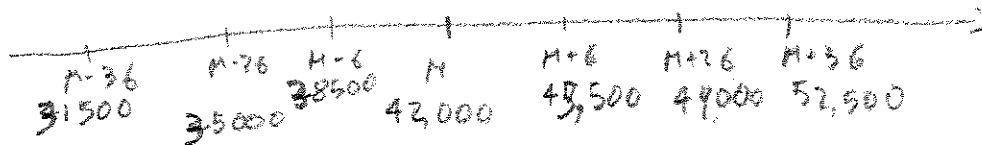
$$s = \sqrt{\frac{SS_x}{n-1}} = \sqrt{\frac{1948}{49}} = 6.31$$

Mean 42,200 \$Standard deviation 6,310 \$

2. (15 pts) Construct an ogive (cumulative frequency graph) for the data given in the previous exercise.



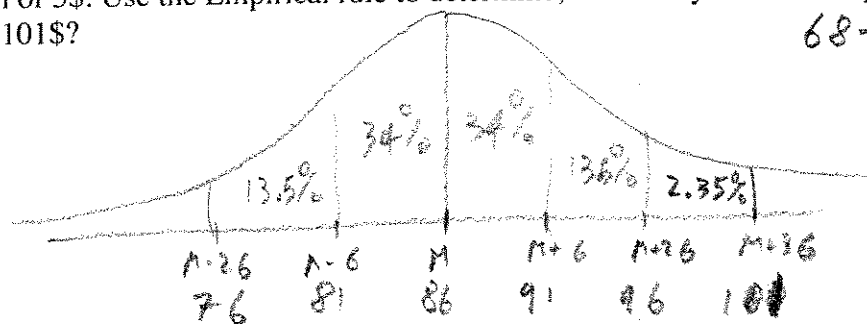
3. (12 pts) The mean employee salary at a given company is \$ 42,000 with a standard deviation of \$ 3,500. If the company employs 450 people, use Chebychev's Theorem to determine about how many employees earn between \$ 31,500 and \$ 52,500?



$$1 - \frac{1}{3^2} = 1 - \frac{1}{9} = 88.9\%$$

4. (15 pts) In a sample of 100 restaurant customers, the mean bill was 86\$ with a standard deviation of 5\$. Use the Empirical rule to determine, how many customers spent between 76\$ and 101\$?

68-95-99.7



$$95 + 2.35 = 97.35\%$$

5. (20 pts) For the stem-and-leaf plot below, find the range, mean, median, mode. (note: for example, 12 | 4 means 124)

10	0 4 7
11	8 8 5 9 9 7
12	8 6 3 1 5
13	3 1 4 7 8 0 9
14	9 6 3
15	2 0

10	0 4 7
11	5 7 8 8 9 9
12	1 3 5 6 8
13	0 1 3 4 7 8 9
14	3 6 9
15	0 2

$N = 26$

$$\text{Mean } \frac{100 + 104 + 107 + \dots + 150 + 152}{26} = \frac{3322}{26} = 127.82$$

$$\text{Range } 152 - 100 = 52$$

$$\text{Median } \frac{126 + 128}{2} = 127$$

$$\text{Mode } 118 \text{ \& } 119$$

Range 52
Mean 127.8
Median 127
Mode 118 \& 119

Answer _____

6. (8 pts) Determine whether the data are qualitative or quantitative:

- a) the colors of automobiles on a used car lot Qualitative
- b) the numbers on the shirts of the football team Qualitative
- c) the number of seats in this class Quantitative
- d) a list of house numbers on your street Qualitative

7. (10 pts) Identify the data set's level of measurement (nominal, ordinal, interval, ratio):

- a) numbers on the shirts of a girl's soccer team nominal
- b) ages of students in a statistics class Ratio
- c) temperatures of 22 selected refrigerators Interval
- d) number of milligrams of tar in 28 cigarettes Ratio
- e) date of birth of students in a given class Interval

EXAM #1Name KEY
(print)

Total = 100 points

Please show all your work.

1. (20 pts) Find the approximate mean and standard deviation of the data set.

Sample annual salaries (in thousand of dollars) for private elementary school teachers:

Salary	Frequency	MIDPOINT	CUM FREQ	$x - M$	$(x - M)^2$
25-29	8	27	8	-10.1	102.01
30-34	11	32	19	-5.1	26.01
35-39	10	37	29	-0.1	.01
40-44	14	42	43	4.9	24.01
45-49	7	47	50	9.9	98.01

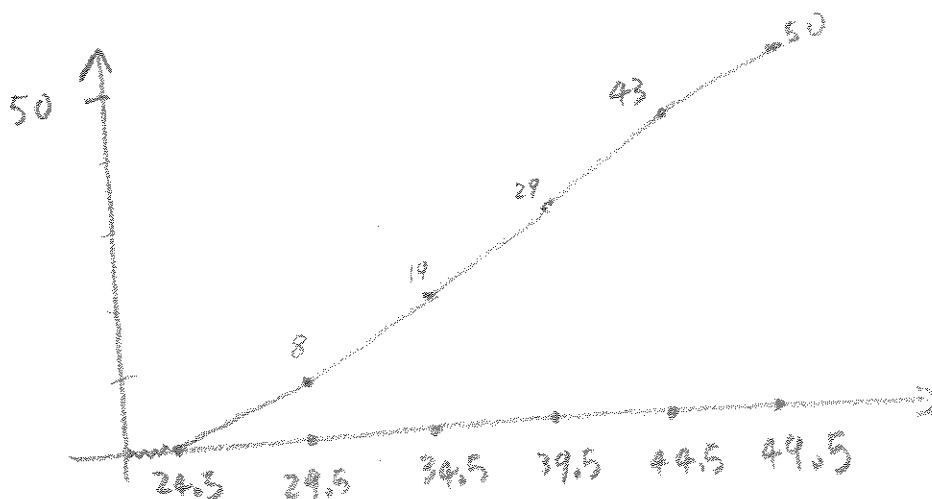
$$M = \frac{8 \times 27 + 11 \times 32 + 10 \times 37 + 14 \times 42 + 7 \times 47}{50} = \frac{1855}{50} = 37.1$$

$$SS_x = 8 \times 102.01 + 11 \times 26.01 + 10 \times .01 + 14 \times 24.01 + 7 \times 98.01 = 2124.5$$

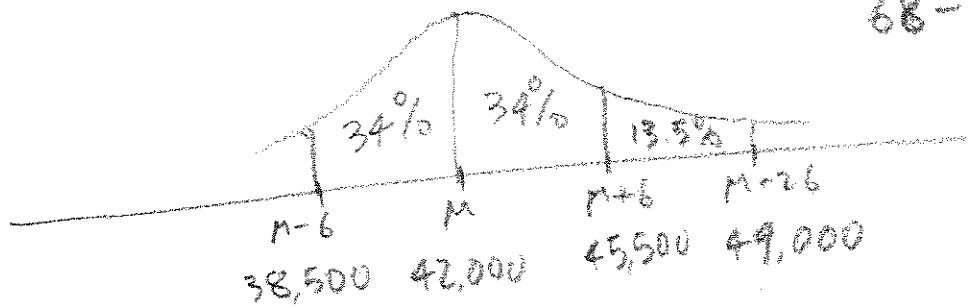
$$s = \sqrt{\frac{SS_x}{n-1}} = \sqrt{\frac{2124.5}{49}} = 6.6$$

Mean 37,100 \$Standard deviation 6,600 \$

2. (15 pts) Construct an ogive (cumulative frequency graph) for the data given in the previous exercise.



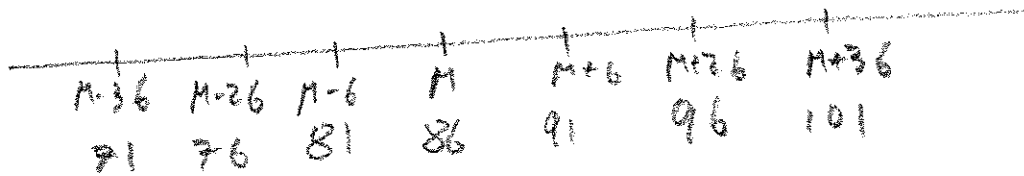
3. (15 pts) The mean employee salary at a given company is \$ 42,000 with a standard deviation of \$ 3,500. If the company employs 450 people, use the Empirical Rule to determine about how many employees earn between \$ 38,500 and \$ 49,000?



$$68 - 95 - 99.7$$

$$68 + 13.5 = 81.5\%$$

4. (12 pts) In a sample of 100 restaurant customers, the mean bill was 86\$ with a standard deviation of 5\$. On the basis of Chebychev's Theorem, how many customers spent between 71\$ and 101\$?



$$1 - \frac{1}{32} = 1 - \frac{1}{9} = 88.9\%$$

5. (20 pts) For the stem-and-leaf plot below, find the range, mean, median, mode. (note: for example, 12 | 4 means 124)

10	0 4 7
11	8 8 5 9 9 7
12	8 6 3 1 5
13	3 1 4 7 8 0 9
14	9 6 3
15	2 0

10	0 4 7
11	5 7 8 8 9 9
12	1 3 5 6 8
13	0 1 3 4 7 8 9
14	3 6 9
15	0 2

N = 26

$$\text{Mean} = \frac{100 + 104 + 107 + \dots + 150 + 152}{26} = \frac{3322}{26} = 127.8$$

$$\text{Range} = 152 - 100 = 52$$

$$\text{Median} = \frac{126 + 128}{2} = 127$$

$$\text{Mode} = 118 \text{ \& } 119$$

$$\text{Range} = 52$$

$$\text{Mean} = 127.8$$

$$\text{Median} = 127$$

$$\text{Mode} = 118 \text{ \& } 119$$

Answer _____

6. (8 pts) Determine whether the data are qualitative or quantitative:

- a) the colors of automobiles on a used car lot Qualitative
- b) the numbers on the shirts of the football team Qualitative
- c) the number of seats in this class Quantitative
- d) a list of house numbers on your street Qualitative

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